UNIT 7 Oscillations

7.1 Defining Simple Harmonic Motion (SHM)

7.2 Frequency and Period of SHM

7.3 Representing and Analyzing SHM

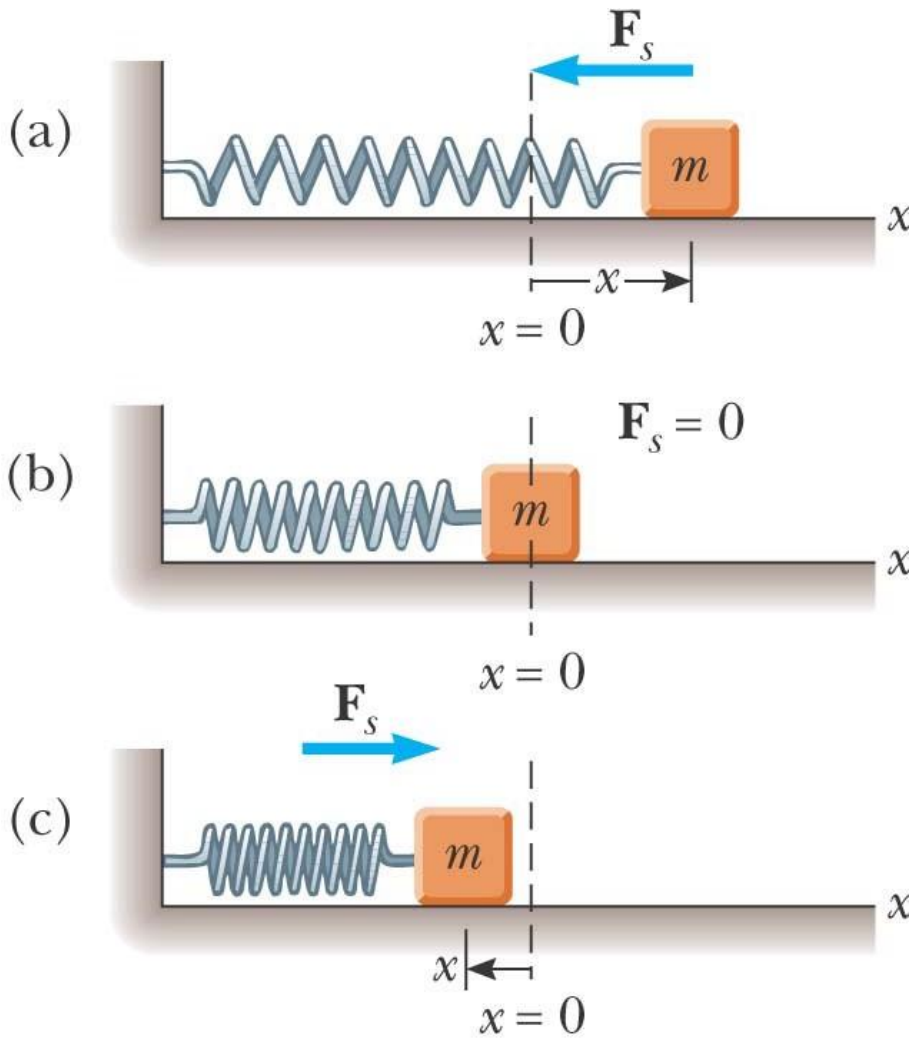
7.4 Energy of Simple Harmonic Oscillators

7.1 Defining Simple Harmonic Motion (SHM)

$$F_s = -kx$$

(Hooke's Law)

F_s is a *restoring force* because it always points toward the equilibrium position (x = 0)



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Characteristic features of simple harmonic motion

- Simple harmonic motion is defined as oscillating motion in which the acceleration is
  - Proportional to the displacement, and
  - Always in the opposite direction to the displacement

$$\boldsymbol{acceleration \propto -displacement}$$

$$or$$

$$\boldsymbol{a \propto -x}$$

In other words, $acceleration\ a = -constant \times displacement$

Applying Newton's Second Law :

$$\Sigma F = ma_x$$

$$-kx = ma_x$$

$$\boxed{a_x = -\frac{k}{m}x}$$

<u>Simple Harmonic Motion</u>

An object moves with *simple harmonic motion* whenever its acceleration is proportional to its position and is *oppositely* directed to the displacement from equilibrium.

Since $a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$,

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

Notice that the acceleration of the particle in SHM is *not constant*. It varies with position x.

If we call $k/m = \omega^2$,

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

(second-order differential equation)

We need a function x(t) whose second derivative is the same as the original function with a negative sign and multiplied by ω^2.

One possible solution is :

$$\boxed{x(t) = A\cos(\omega t + \phi)}$$

$$x(t) = A\cos(\omega t + \phi)$$

<u>Amplitude (A)</u> : maximum value of the position of the particle in either the positive or negative direction.

<u>Angular Frequency (ω)</u> : number of oscillations per second.

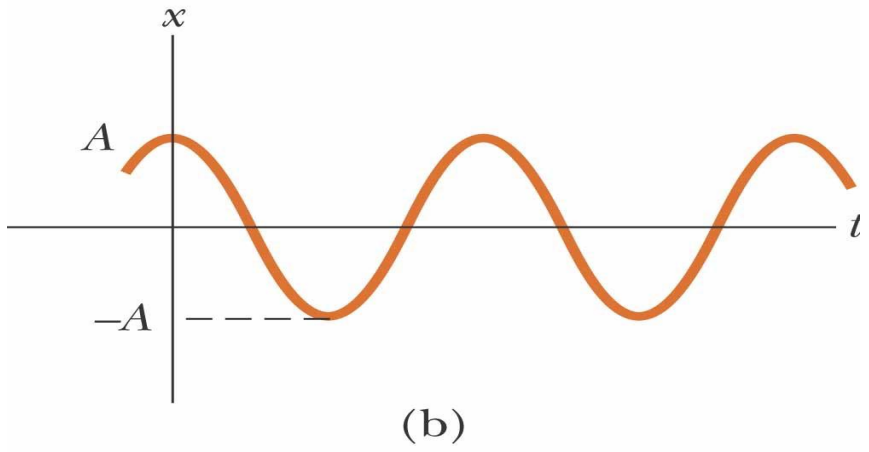
Since $k/m = \omega^2$,

$$\omega = \sqrt{\frac{k}{m}}$$

<u>Phase Constant (f)</u> : initial phase angle.

This is determined by the position of the particle at t = 0.

If particle is at maximum position x = A at t = 0, the phase constant f = 0.



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f = 0

7.2 Frequency and Period of SHM

<u>Frequency (f)</u> : the inverse of period. The number of oscillations that the particle undergoes per second.

$$\boxed{f = \frac{1}{T}}$$

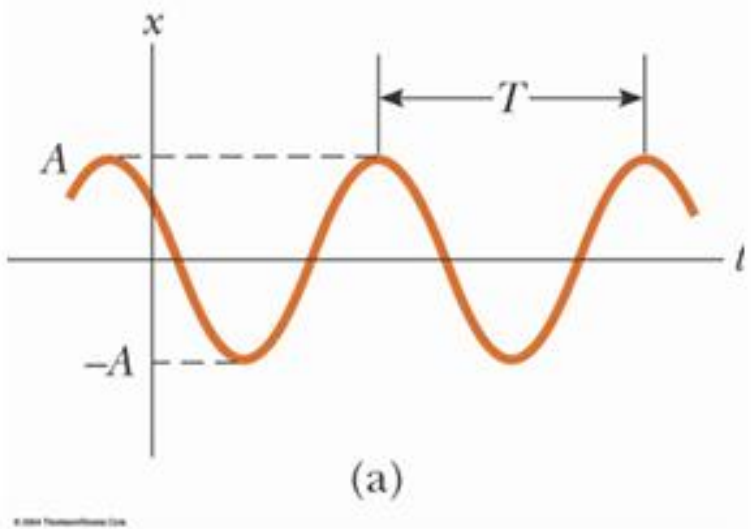
Since $T = 2\pi/\omega$,

$$\boxed{f = \frac{\omega}{2\pi}}$$

The Hertz (Hz) is the SI unit for frequency.

<u>Period (T)</u> : the time interval required for the particle to go through one full cycle of its motion.

$$T = \frac{2\pi}{\omega}$$



Since $f = \omega/2\pi$,

$$\boxed{\omega = 2\pi f = \frac{2\pi}{T}}$$

Since $\omega = \sqrt{k/m}$ and $T = 2\pi/\omega$,

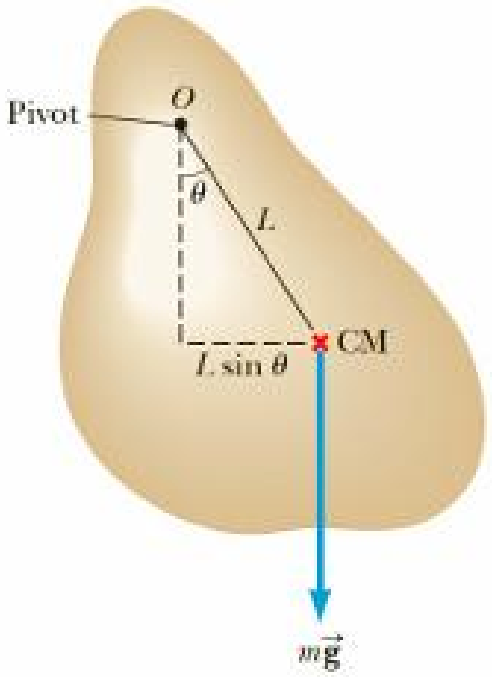
$$\boxed{T = 2\pi \sqrt{\frac{m}{k}}}$$

depend only on mass and spring constant

$$\boxed{f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}}$$

The Physical Pendulum

- The simple pendulum discussed thus far consists of a mass attached to a string.
- A pendulum, however, can be made from an object of any shape. The general case is called the physical pendulum as shown below.
  - **In this figure a rigid object is pivoted at point O, which is a distance L from the object's center of mass.**
  - **The center of mass oscillates along a circular arc, just like the simple pendulum.**



A physical pendulum pivoted at O.

The Physical Pendulum

- The period of a physical pendulum is given by

$$\boxed{T = 2\pi \sqrt{\frac{I}{mgL}}}$$

where **I** is the object's moment of inertia and m is the object's mass.

Example 15.6 A Swinging Rod

A uniform rod of mass M and length L is pivoted about one end and oscillates in a vertical plane (Fig. 15.18). Find the period of oscillation if the amplitude of the motion is small.

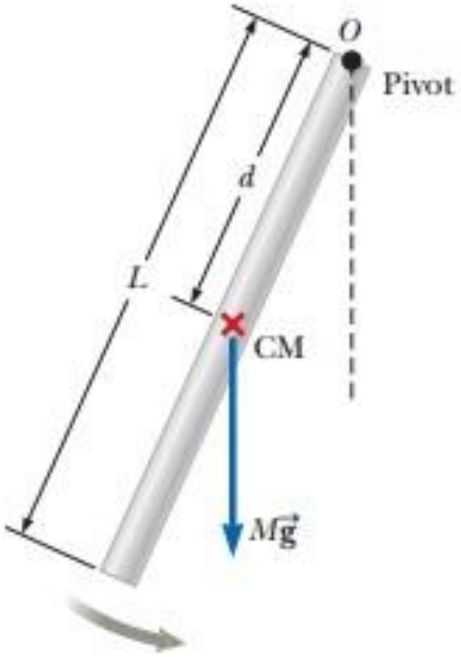


Figure 15.18 (Example 15.6) A rigid rod oscillating about a pivot through one end is a physical pendulum with $d = L/2$.

7.3 Representing and Analyzing SHM

Velocity in Simple Harmonic Motion

$$\frac{dx}{dt} = A \frac{d}{dt} \cos(\omega t + \phi) = -\omega A \sin(\omega t + \phi)$$

$$\boxed{v = -\omega A \sin(\omega t + \phi)}$$

Acceleration in Simple Harmonic Motion

$$\frac{d^2x}{dt^2} = -\omega A \frac{d}{dt} \sin(\omega t + \phi) = -\omega^2 A \cos(\omega t + \phi)$$

$$\boxed{a = -\omega^2 A \cos(\omega t + \phi)}$$

Maximum Speed in Simple Harmonic Motion

Since sine and cosine oscillate between +/- 1,

$$v_{max} = +/- \omega A$$

$$\boxed{v_{max} = \sqrt{\frac{k}{m}}\ A}$$ (magnitude only)

Maximum Acc. in Simple Harmonic Motion

$$a_{max} = +/- \omega^2 A$$

$$\boxed{a_{max} = \frac{k}{m}\ A}$$ (magnitude only)

Example 15.1 A Block–Spring System AM

A 200-g block connected to a light spring for which the force constant is 5.00 N/m is free to oscillate on a frictionless, horizontal surface. The block is displaced 5.00 cm from equilibrium and released from rest as in Figure 15.6.

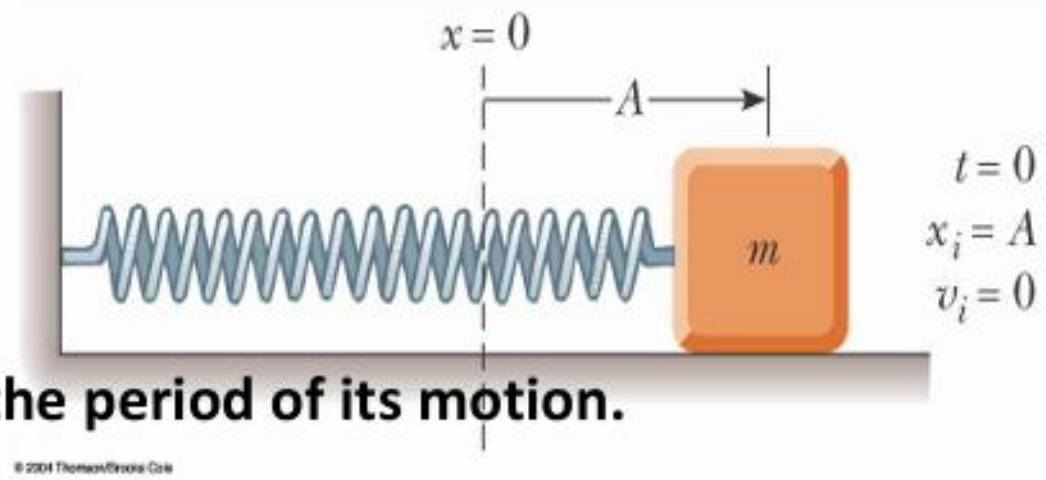
(A) Find the period of its motion.

(B) Determine the maximum speed of the block.

(C) What is the maximum acceleration of the block?

(D) Express the position, velocity, and acceleration as functions of time in SI units.

A 200 g block connected to a light spring for which the force constant is 5.00 N/m is free to oscillate on a horizontal, frictionless surface. The block is displaced 5.00 cm from equilibrium and released from rest, as shown below.



a) Find the period of its motion.

b) Determine the maximum speed of the block.

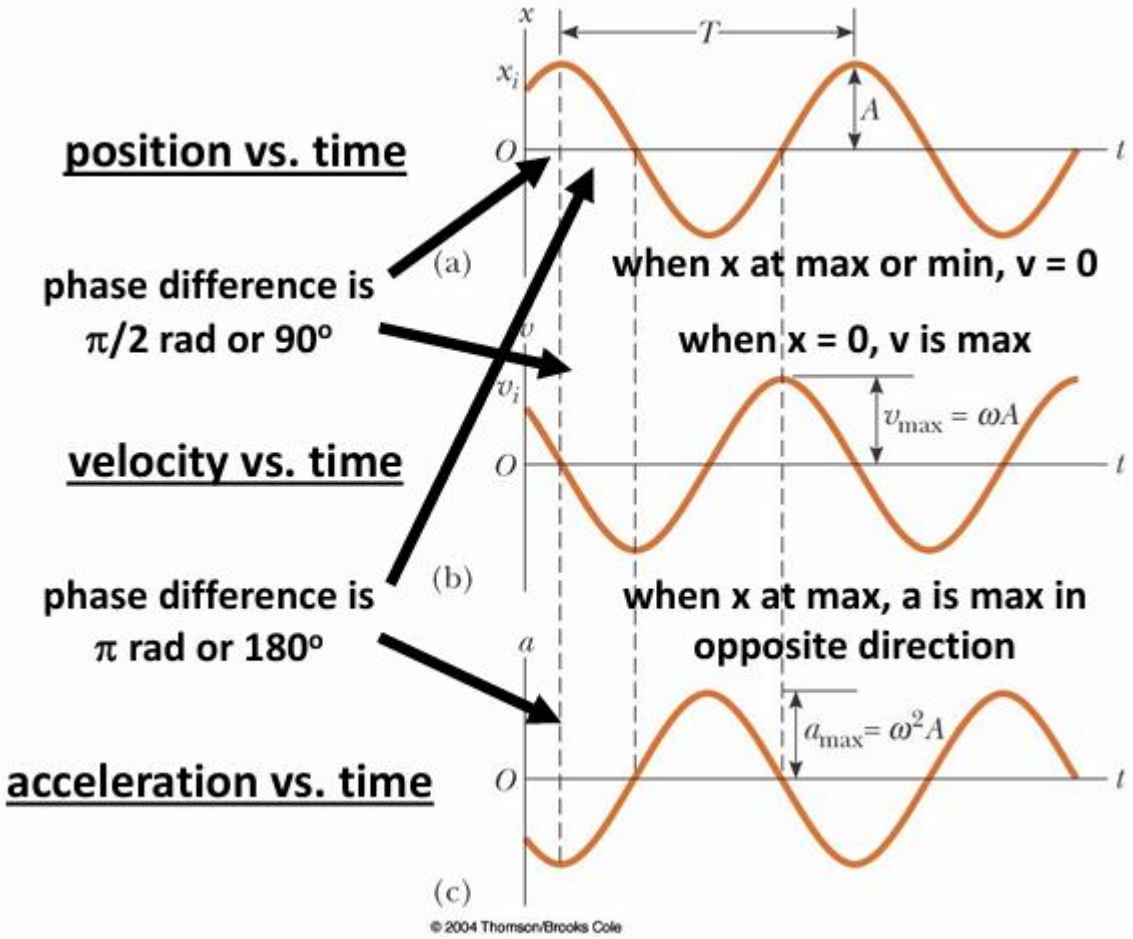
c) What is the maximum acceleration of the block ?

d) Express the position, speed, and acceleration as functions of time.

Example 15.2 Watch Out for Potholes! AM

A car with a mass of 1 300 kg is constructed so that its frame is supported by four springs. Each spring has a force constant of 20 000 N/m. Two people riding in the car have a combined mass of 160 kg. Find the frequency of vibration of the car after it is driven over a pothole in the road.

continued



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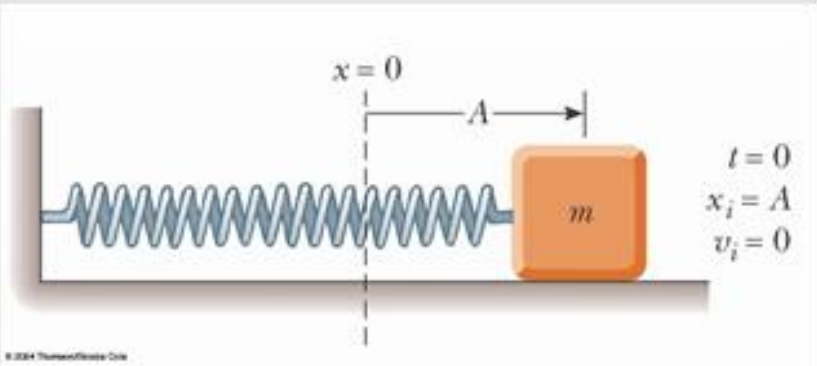
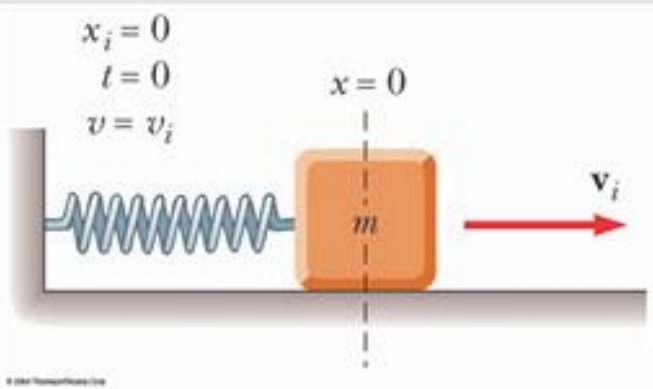
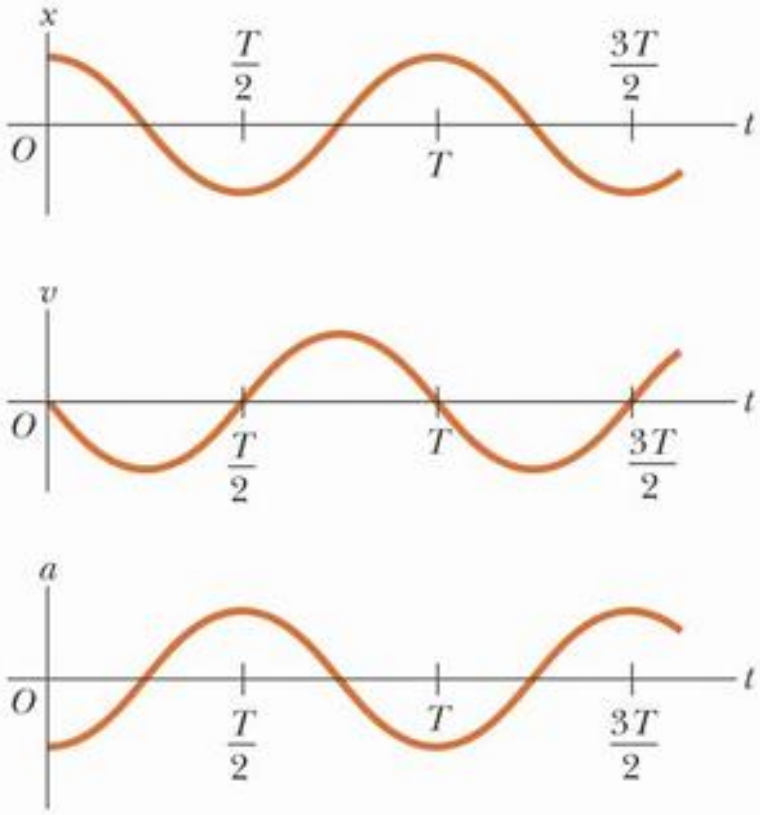

x = 0
A
m
t = 0
x_i = A
v_i = 0


x_i = 0
t = 0
v = v_i
x = 0
m
v_i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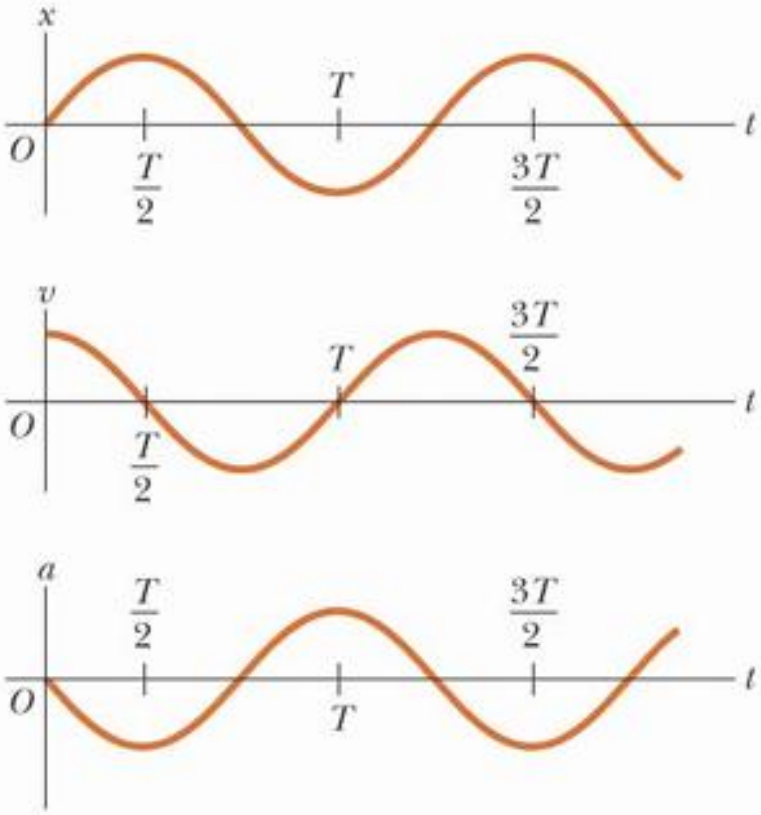
t
T/2
T
3T/2


x
v
a
O
t
T/2
T
3T/2

7.4 Energy of Simple Harmonic Oscillators

KE of Block

$$KE = \frac{1}{2} mv^2$$

Since $v = -\omega A \sin(\omega t + \phi)$,

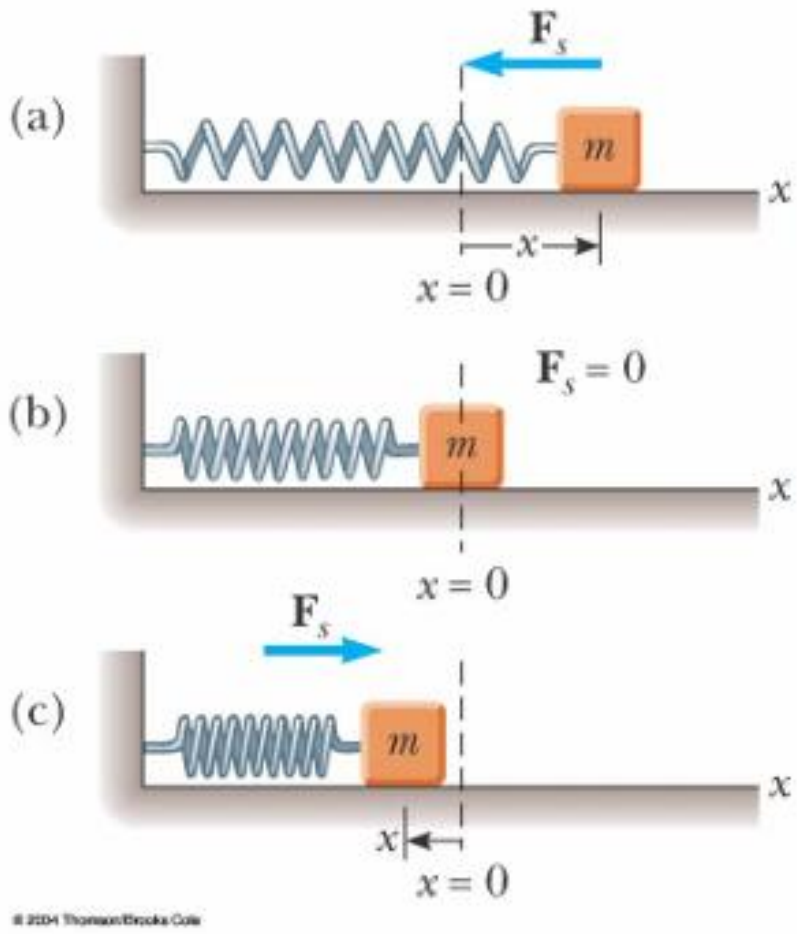
$$KE = \frac{1}{2} m\omega^2 A^2 \sin^2(\omega t + \phi)$$

Elastic PE Energy Stored in Spring

$$U = \frac{1}{2} kx^2$$

Since $x = A \cos(\omega t + \phi)$,

$$U = \frac{1}{2} kA^2 \cos^2(\omega t + \phi)$$



Total Mechanical Energy of Simple Harmonic Oscillator

$$E = KE + U$$

$$E = \frac{1}{2} m\omega^2 A^2 \sin^2(\omega t + \phi) + \frac{1}{2} kA^2 \cos^2(\omega t + \phi)$$

Since $\omega^2 = k/m$,

$$E = \frac{1}{2} kA^2 [\sin^2(\omega t + \phi) + \cos^2(\omega t + \phi)]$$

Since $\sin^2\theta + \cos^2\theta = 1$,

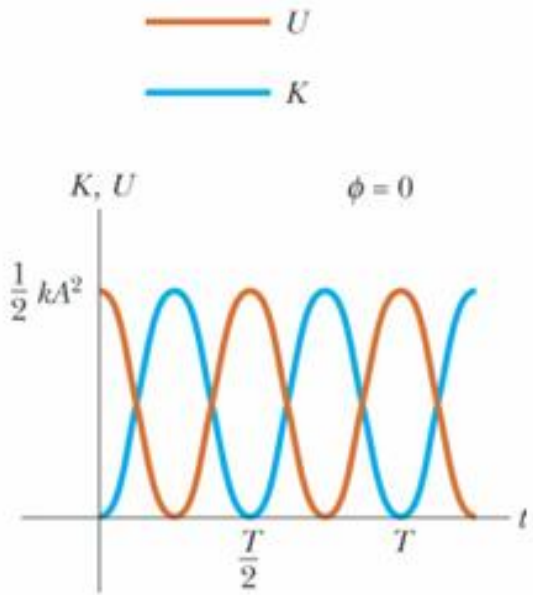
$$\boxed{E = \frac{1}{2} kA^2}$$

$$E = \frac{1}{2} kA^2$$

The total mechanical energy of a simple harmonic oscillator is a constant of the motion and is proportional to the *square* of the amplitude.

U is small when KE is large, and vice versa.

KE + U = constant



$$E = \frac{1}{2} kA^2$$

Since E = KE + U,

$$\frac{1}{2} kA^2 = \frac{1}{2} mv^2 + \frac{1}{2} kx^2$$

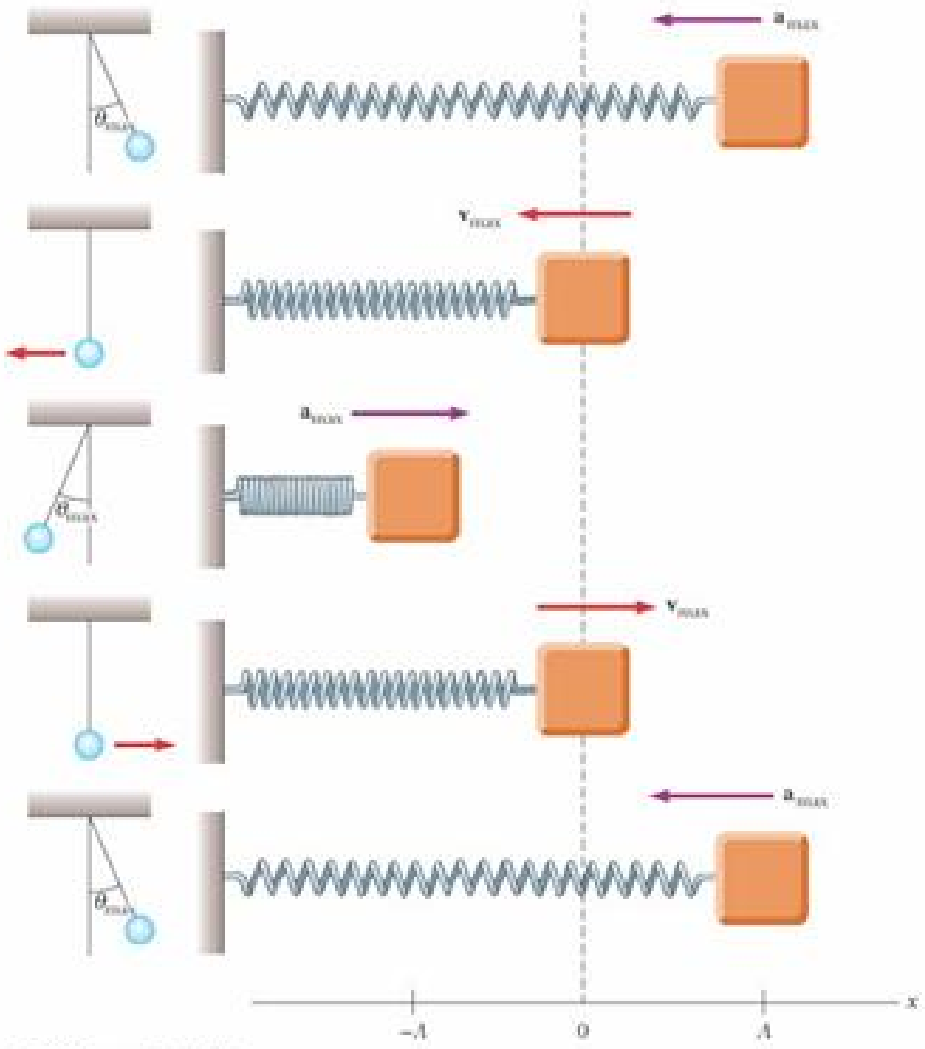
Solving for v,

$$v = +/- \sqrt{\frac{k}{m} (A^2 - x^2)}$$

Since $\omega = \sqrt{\frac{k}{m}}$,

$$\boxed{v = +/- \omega \sqrt{A^2 - x^2}}$$

velocity of simple harmonic oscillator



t	x	v	a	K	U
0	A	0	$-\omega^2 A$	0	$\frac{1}{2}kA^2$
$T/4$	0	$-\omega A$	0	$\frac{1}{2}kA^2$	0
$T/2$	$-A$	0	$\omega^2 A$	0	$\frac{1}{2}kA^2$
$3T/4$	0	ωA	0	$\frac{1}{2}kA^2$	0
T	A	0	$-\omega^2 A$	0	$\frac{1}{2}kA^2$

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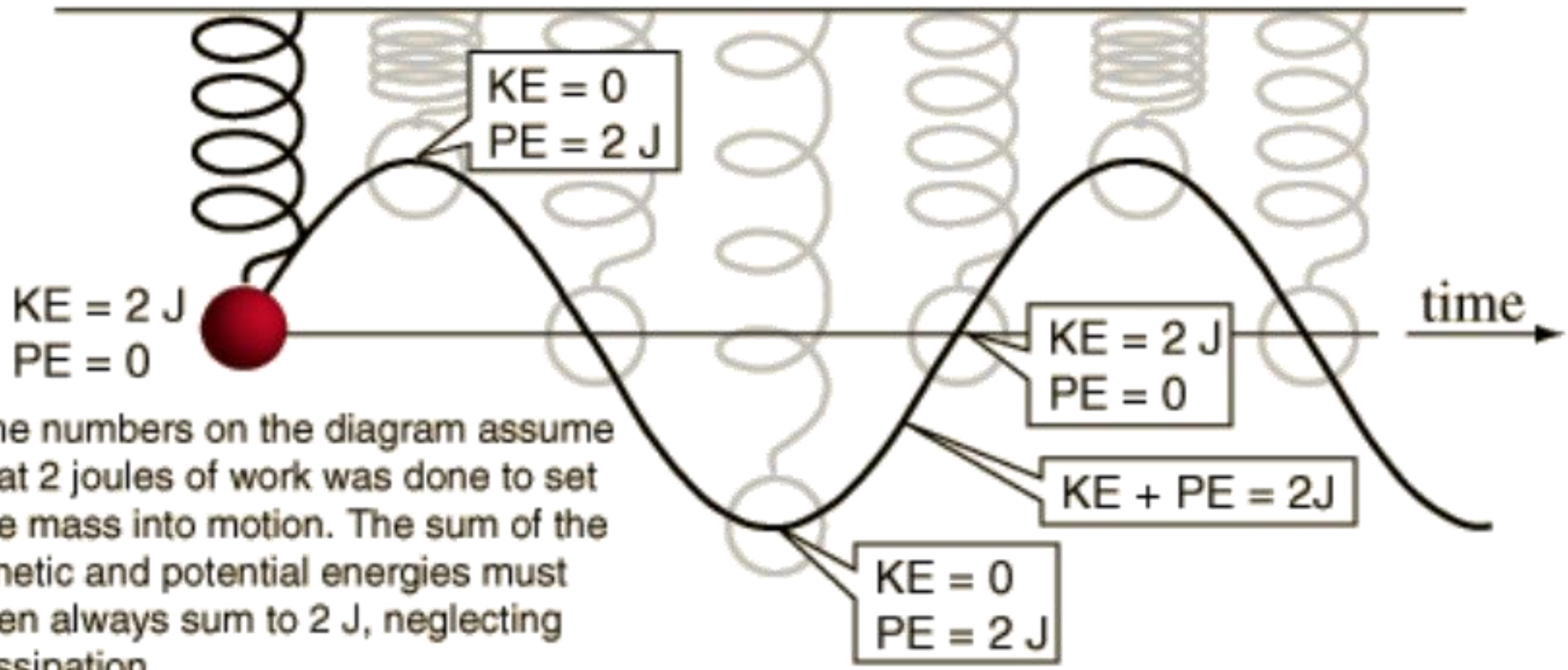
A 0.500 kg cart connected to a light spring for which the force constant is 20.0 N/m oscillates on a horizontal, frictionless air track.

a) Calculate the total energy of the system and the maximum speed of the cart if the amplitude of the motion is 3.00 cm.

b) What is the velocity of the cart when the position is 2.00 cm ?

c) Compare the kinetic and potential energies of the system when the position is 2.00 cm ?



The numbers on the diagram assume that 2 joules of work was done to set the mass into motion. The sum of the kinetic and potential energies must then always sum to 2 J, neglecting dissipation.

Example

A slingshot consists of a light leather cup, containing a stone, that is pulled back against 2 rubber bands. It takes a force of 30 N to stretch the bands 1.0 cm (a) What is the potential energy stored in the bands when a 50.0 g stone is placed in the cup and pulled back 0.20 m from the equilibrium position? (b) With what speed does it leave the slingshot?

$a)\ F_s = kx \quad 30 = k(0.01) \quad k =$ **3000 N/m**

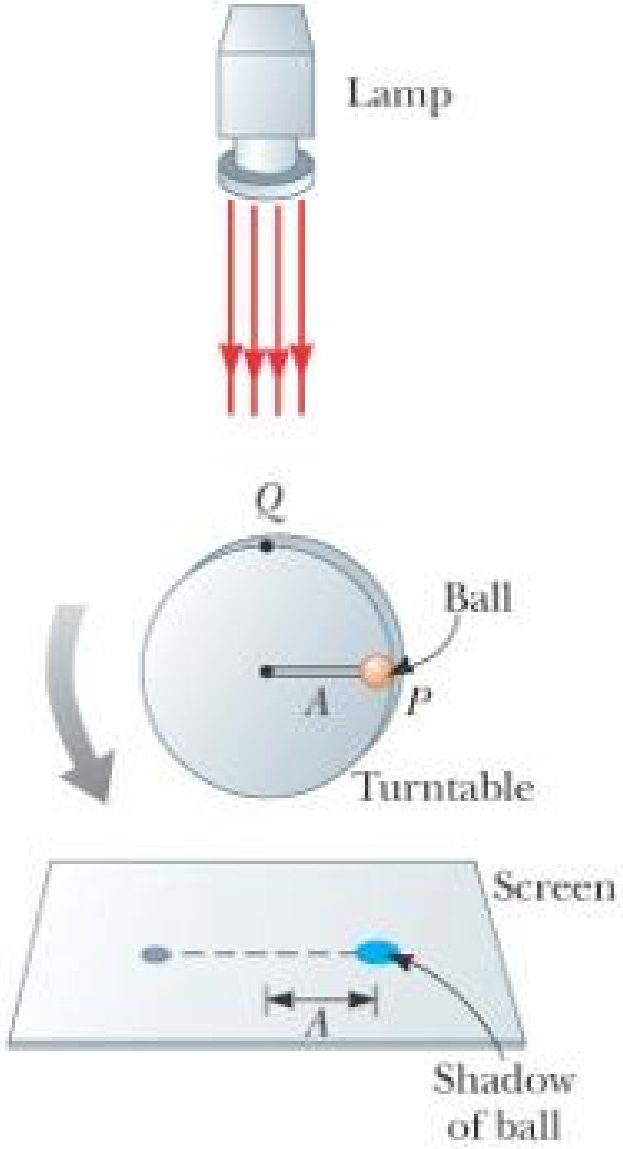
$U_s = \frac{1}{2}kx^2 = 0.5(k)(.20)^2 =$ **60 J**

$b)\ E_B = E_A \quad U_s = K$

$U_s = \frac{1}{2}mv^2 = \frac{1}{2}(0.050)v^2$

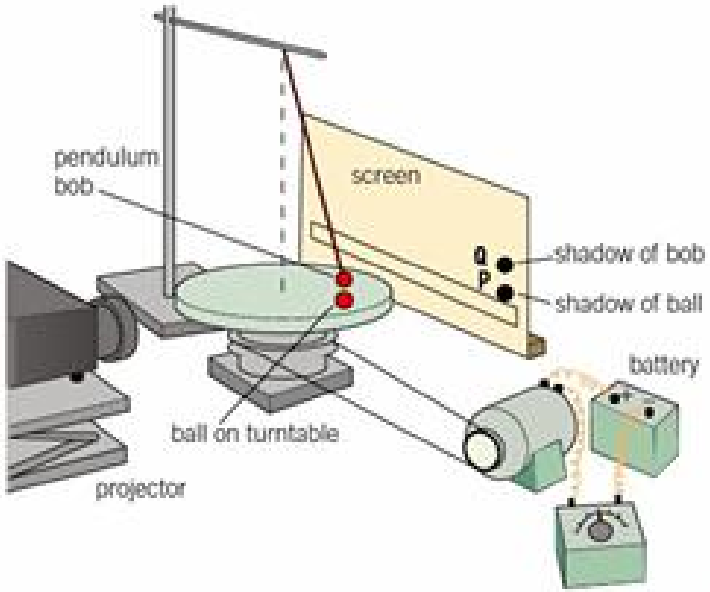
$v =$ **49 m/s**

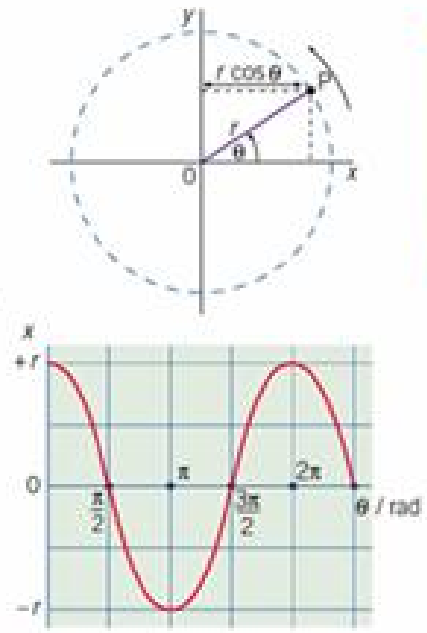
As the turntable rotates with constant angular speed, the shadow of the ball moves back and forth in simple harmonic motion.



Comparing simple harmonic motion with circular motion

- Consider motion of a ball and the pendulum bob in the figure below:





▲ **Figure 1** *Circles and waves*

$$v = \frac{2\pi r}{T}, \quad r = A$$

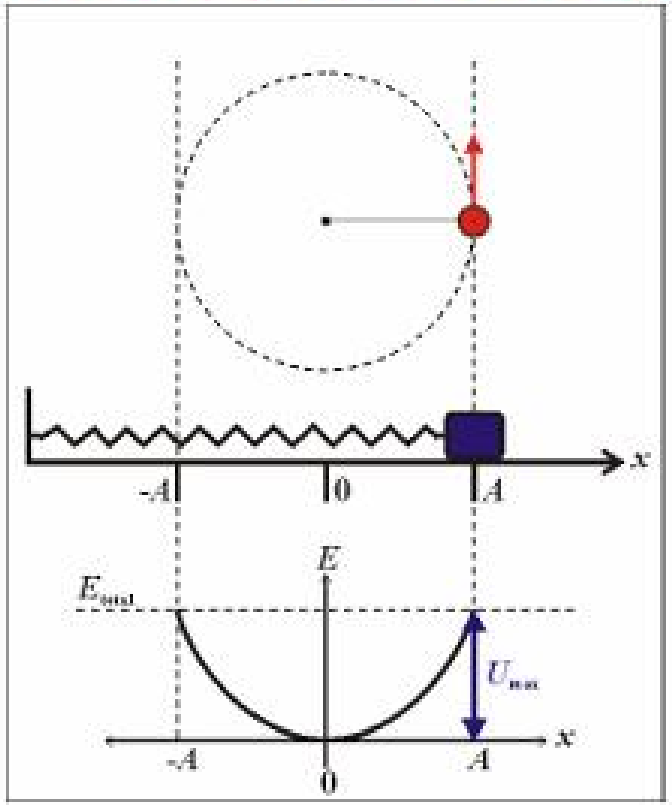
$$v = \frac{2\pi A}{T} \rightarrow \frac{T}{2\pi} = \frac{A}{v}$$

SHM and Uniform Circular Motion

Springs and Waves behave very similar to objects that move in circles.

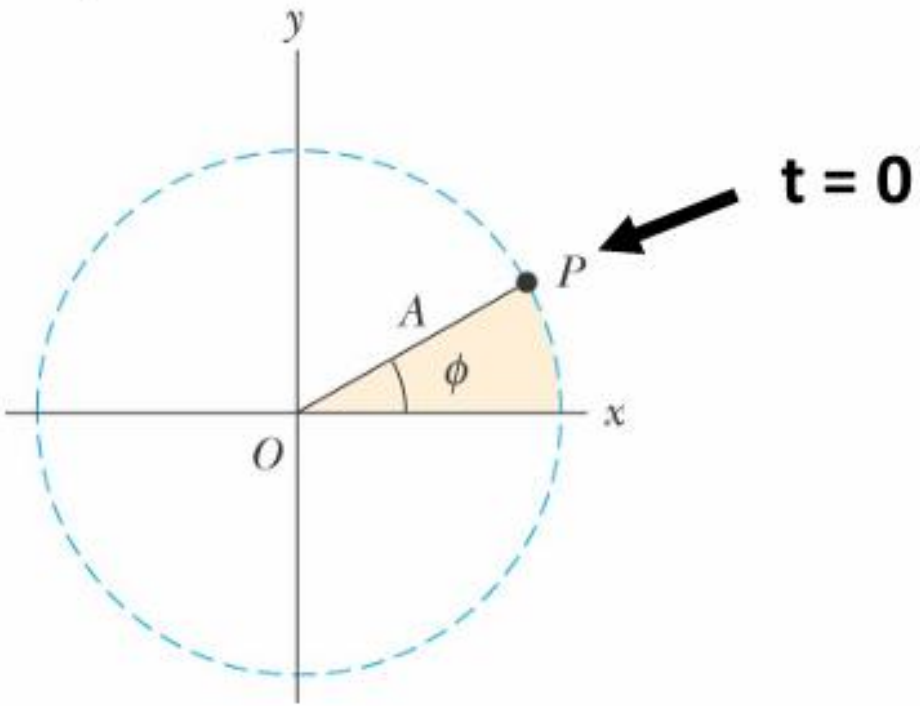
The radius of the circle is symbolic of the displacement, x, of a spring or the amplitude, A, of a wave.

$$x_{spring} = A_{wave} = r_{circle}$$



Reference Circle

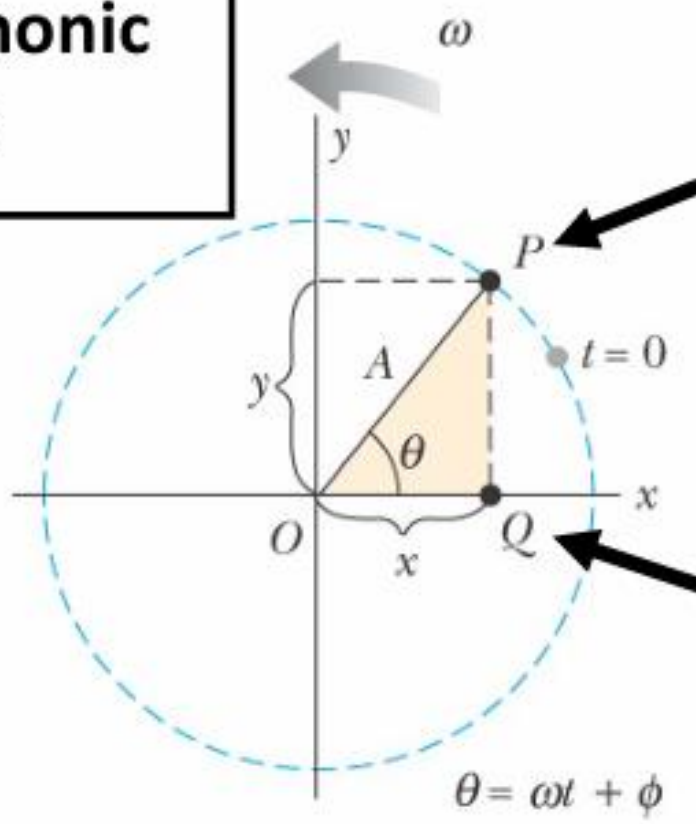
Simple harmonic motion along a straight line can be represented by the projection of uniform circular motion along a diameter of a *reference circle*.



Since $\theta = \omega t = \phi$,

x coordinate is $x(t) = A\cos(\omega t + \phi)$

Point Q moves with simple harmonic motion



t > 0

Q is the projection of P

Energy of a Simple Harmonic Oscillator

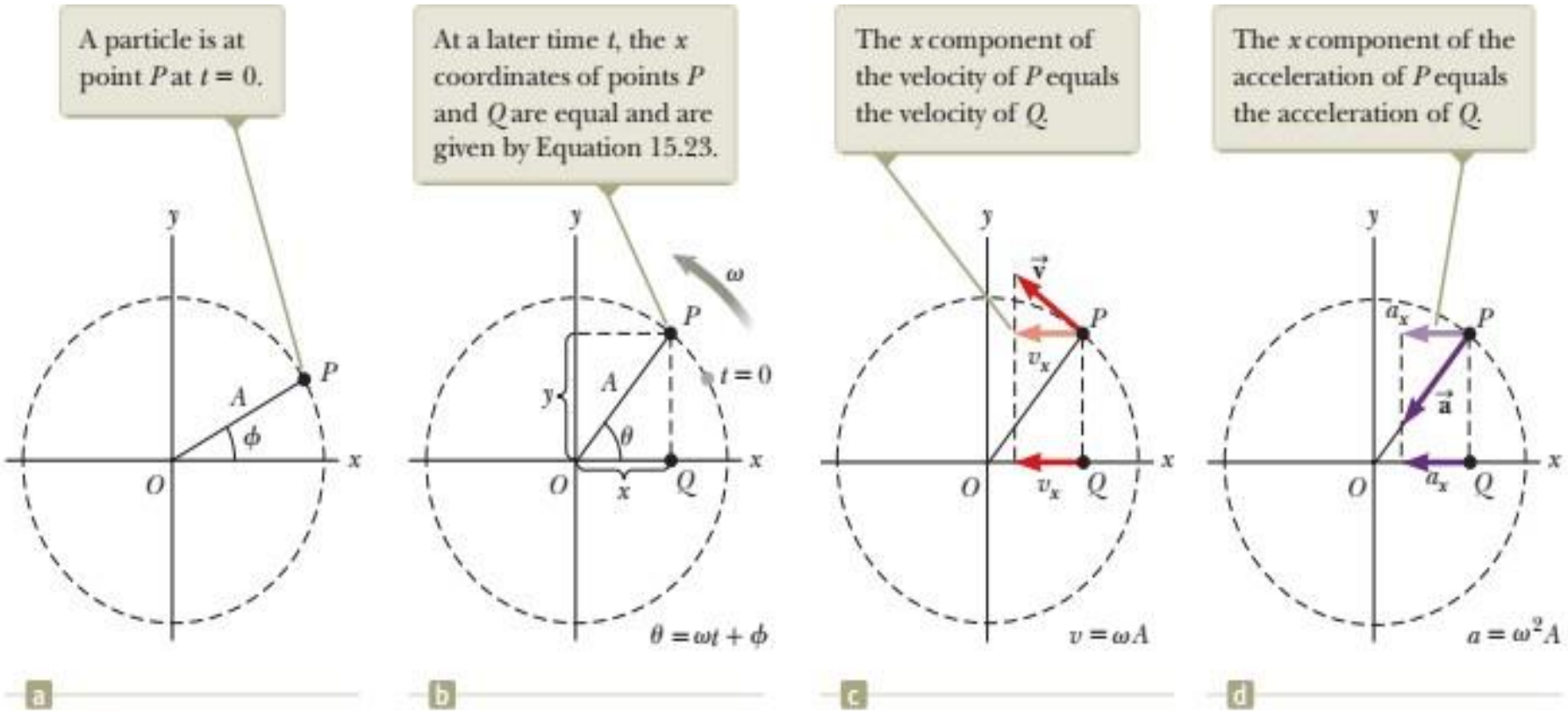


Figure 15.14 Relationship between the uniform circular motion of a point P and the simple harmonic motion of a point Q. A particle at P moves in a circle of radius A with constant angular speed ω.

Energy of a Simple Harmonic Oscillator

Example 15.4 **Circular Motion with Constant Angular Speed** AM

The ball in Figure 15.13 rotates counterclockwise in a circle of radius 3.00 m with a constant angular speed of 8.00 rad/s. At $t = 0$, its shadow has an x coordinate of 2.00 m and is moving to the right.

(A) Determine the x coordinate of the shadow as a function of time in SI units.

(B) Find the x components of the shadow's velocity and acceleration at any time t.

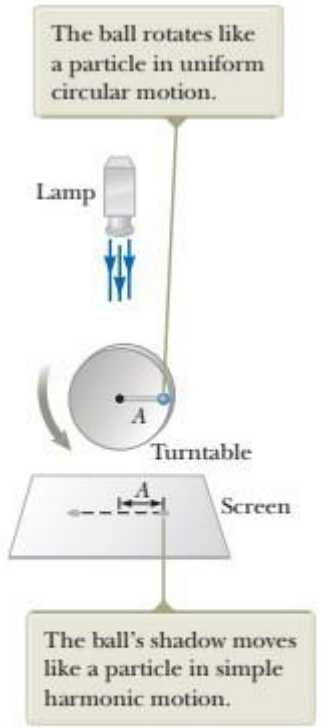
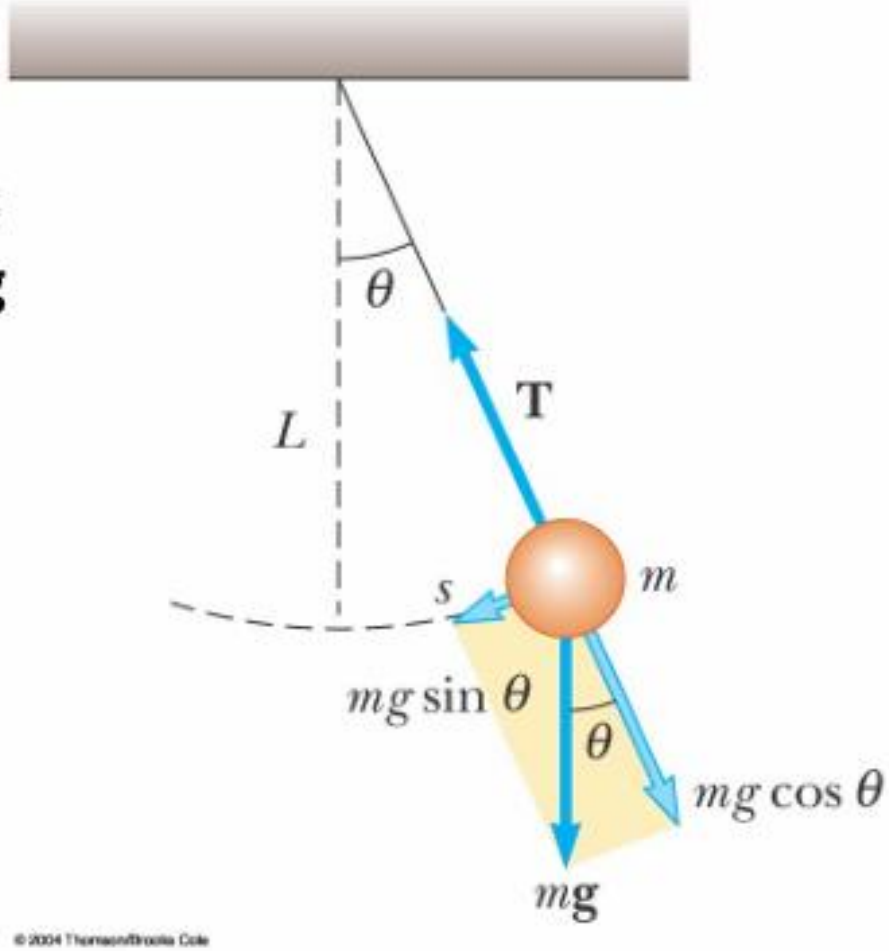


Figure 15.13 An experimental setup for demonstrating the connection between a particle in simple harmonic motion and a corresponding particle in uniform circular motion.

A *simple pendulum* exhibits periodic motion. It consists of a particle-like bob of mass *m* suspended by a light string of length *L* that is fixed at the upper end.



<u>Forces acting on bob</u>

Tension in string

Gravitational force mg

$F_t = mg \sin\theta$ always acts opposite to the displacement of the bob

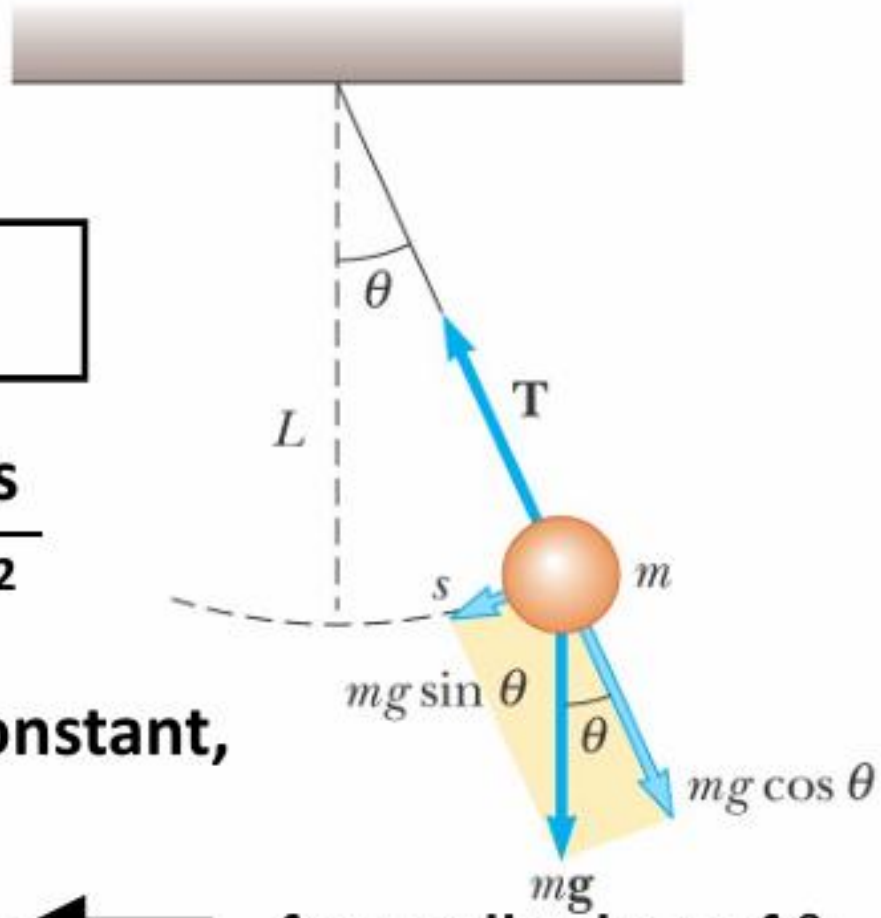
F_t is a restoring force

$$F_t = -mg \sin\theta = m \frac{d^2s}{dt^2}$$

Since $s = L\theta$ and L is constant,

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L} \sin\theta$$

for small values of θ



$$\left.\begin{aligned} \frac{d^2\theta}{dt^2} &= -\frac{g}{L}\sin\theta \\ \frac{d^2x}{dt^2} &= -\frac{k}{m}x \end{aligned}\right\} \text{same form}$$

For small amplitudes (θ < about 10°), the motion of a pendulum is *close* to simple harmonic motion.

Instead of $x = A\cos(\omega t + \phi)$,

$$\boxed{\theta = \theta_{max}\cos(\omega t + \phi)}$$

Instead of $\omega = \sqrt{\dfrac{k}{m}}$,

angular frequency for a simple pendulum →

$$\boxed{\omega = \sqrt{\frac{g}{L}}}$$

$$\boxed{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L}{g}}}$$

$$\omega = \sqrt{\frac{g}{L}} \qquad T = 2\pi\sqrt{\frac{L}{g}}$$

The period and frequency of a simple pendulum depend <u>only</u> on the length of the string and the acceleration due to gravity.

(Period and frequency are independent of mass.)

Example 2

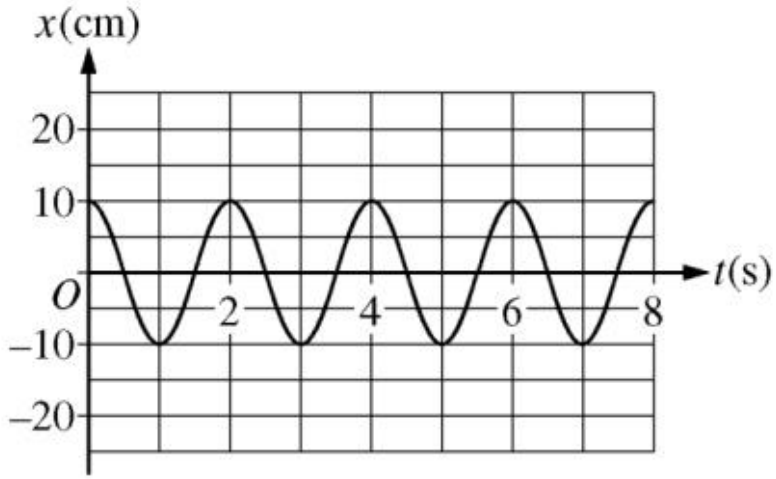
A simple pendulum has a mass of 0.250 kg and a length of 1.00 m. It is displaced through an angle of 15.0° and released. What is the

a) maximum speed

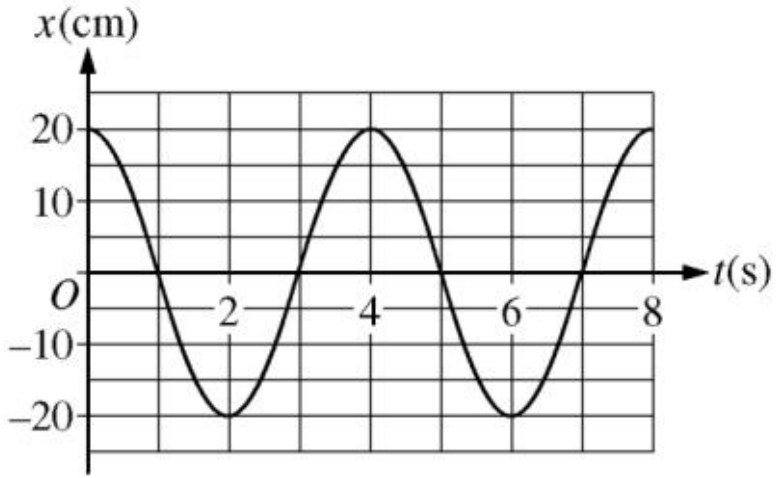
b) maximum angular acceleration

c) maximum restoring force ?

Example Question of AP Classroom



Block *A*



Block *B*

Two blocks are connected to identical ideal springs and are oscillating on a horizontal frictionless surface. Block A has mass m, and its motion is represented by the graph of position as a function of time shown above on the left. Block B's motion is represented above on the right. Which of the following statements comparing block B to block A is correct?

(A) Because block B covers more distance per cycle than block A, block B takes more time to complete each cycle.

(B) Because the spring attached to block B is initially stretched a greater distance, the spring constant is smaller and therefore block B has a slower average speed than block A does.

(C) Because block B has more mass, it has a slower average speed than block A does.

(D) Because block B has more mass, its acceleration is smaller than that of block A at any given displacement from the equilibrium position.

Example Question of AP Classroom

A block attached to the lower end of a vertical spring oscillates up and down. If the spring obeys Hooke's law, the period of oscillation depends on which of the following?

I. Mass of the block

II. Amplitude of the oscillation

III. Force constant of the spring

(A) I only
(B) II only
(C) III only
(D) I and II
(E) I and III

Example Question of AP Classroom

An object moves up and down the y-axis with an acceleration given as a function of time t by the expression $a = A \sin \omega t$, where A and ω are constants. What is the period of this motion?

(A) ω

(B) $2\pi\omega$

(C) $\omega^2 A$

(D) $2\pi/\omega$

(E) $\omega/2\pi$